

# Digital System Design

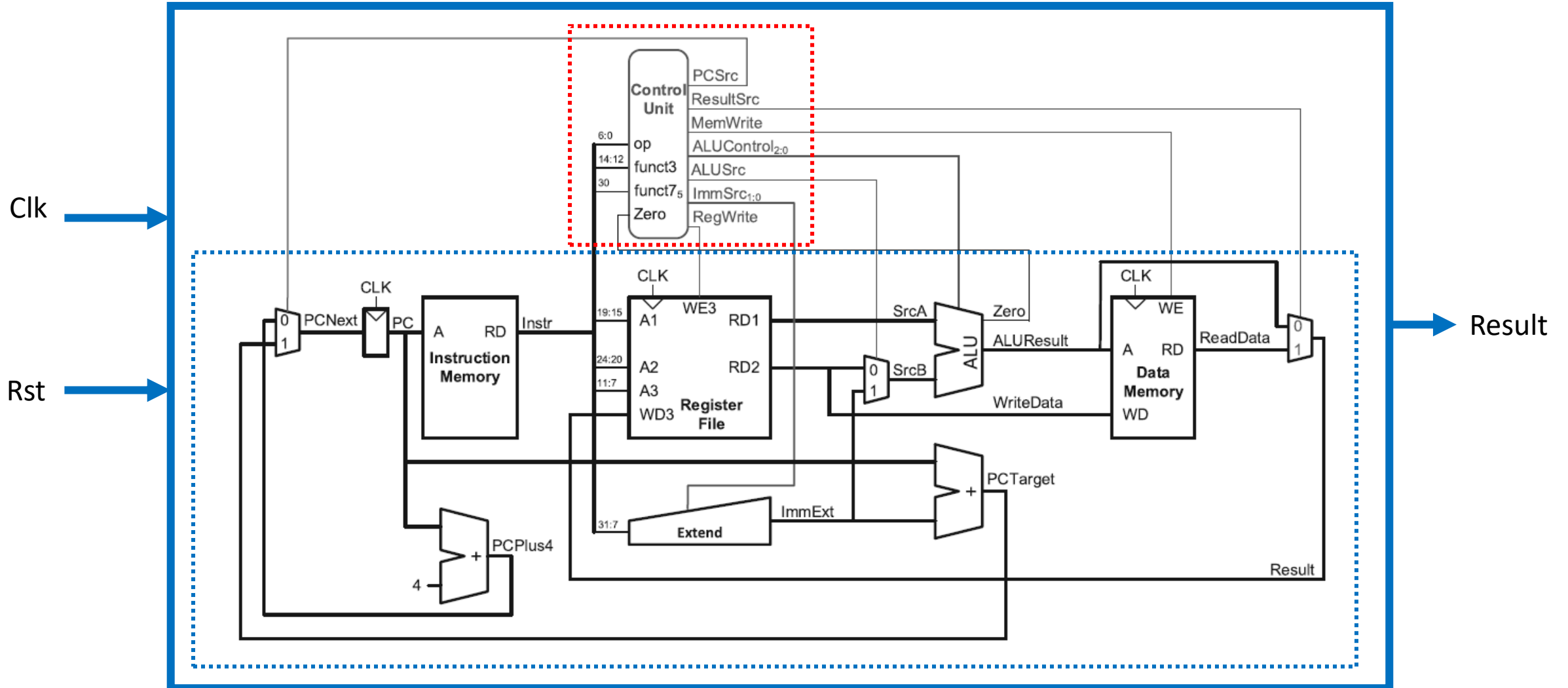
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Dr. Nazanin Ghasemian

Fall 2026 · Lecture 8



# RISC\_V Microarchitecture



# State Elements of a RISC-V Processor

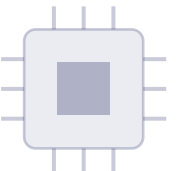
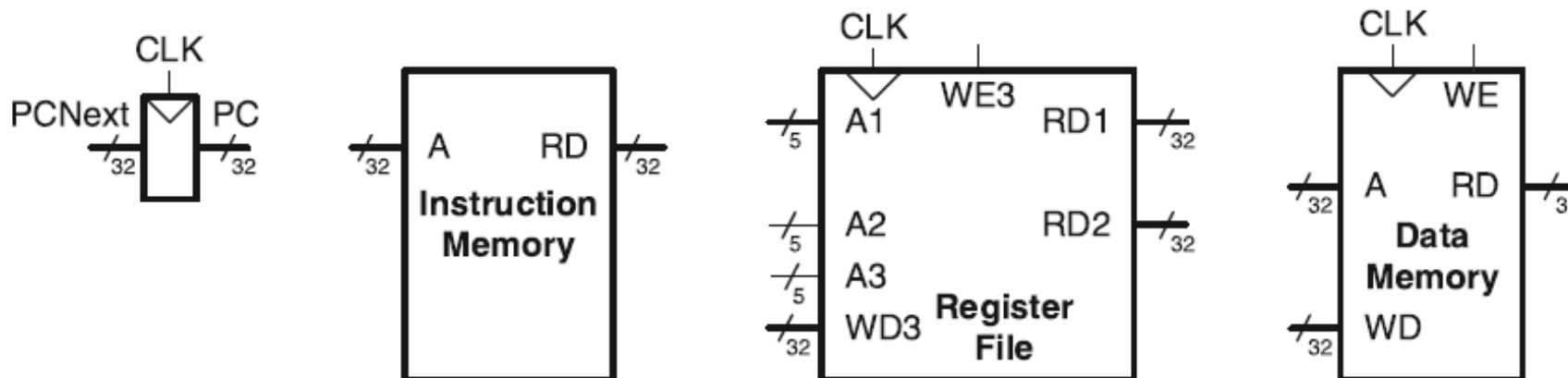
- The initial processor contains four state elements:
  - Program counter
  - Instruction memory
  - Register file
  - Data memory
- The program counter holds the address of the current instruction.

## Register file

Two combinational read ports

One clocked write port

Register x0 always contains zero



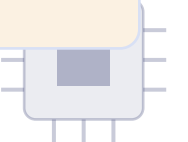
# Reading and Writing State Elements

## 1 Combinational Reads

- Instruction memory, data memory, and the register file produce new outputs when their addresses change.
- Reading does not require a clock edge.

## 2 Synchronous Writes

- The PC, register file, and data memory update on the rising clock edge.
- Input data and control signals must satisfy setup and hold time requirements.



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